Title: Strength of Materials

Author: R.S. Khurmi, N. Khurmi

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Chapter 4: Bending Stress in Beams

When a beam is subjected to a bending moment M, the bending equation sigma / y = M / I = E / R relates the bending stress sigma at a distance y from the neutral axis to the moment of inertia I and the modulus of elasticity E. (rescanned copy, minor OCR noise)

Example 4.3: A simply supported beam of length 4 m carries a point load of 20 kN at mid-span. The cross-section is rectangular, 100 mm wide and 200 mm deep. Determine the maximum bending stress sigma_max in N/mm^2 (MPa), given that the bending moment M = W L / 4.

Solution: M = 20 kN * 4 m / 4 = 20 kN*m. I = b d^3 / 12 = 100 * 200^3 / 12 mm^4. sigma_max = M y / I where y = d / 2 = 100 mm.

Problem 4.12: Repeat the above calculation for a cantilever beam of length 2 m carrying a point load of 10 kN at the free end, with the same rectangular cross-section.

Chapter 5: Shear Force and Torsion

The shear stress tau in a circular shaft of radius R subjected to torque T is tau = T r / J, where J is the polar moment of inertia and r is the radial distance from the shaft axis, measured in N/mm^2.